



Al-Balqa' Applied University
Faculty of Engineering Technology, Department of Physics and Basic Science
Mid Exam, Probability and Random Variables

Name:

Lecture Time:

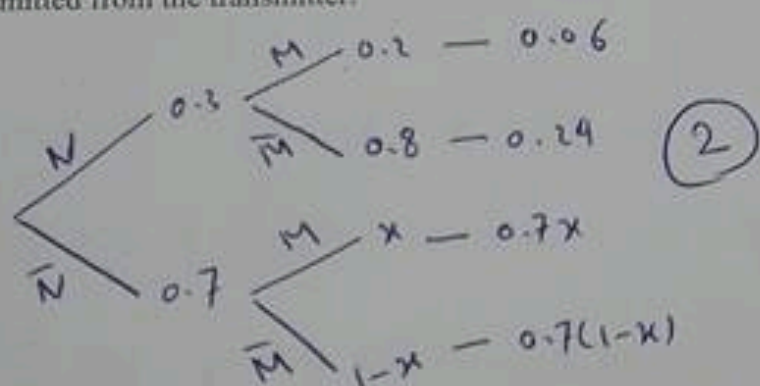
Date: 19/03/2017

Part A: Written questions

Question 1: Find the c.d.f. for the Rayleigh density that is given by $f(x) = \begin{cases} 2\lambda x e^{-\lambda x^2}, & 0 \leq x < \infty \\ 0, & \text{o.w.} \end{cases}$ (5 marks)

$$\begin{aligned} F(x) &= \int_0^x 2\lambda u e^{-\lambda u^2} du \quad (1) \\ &= \int_0^{-\lambda x^2} 2\lambda u e^y \frac{dy}{-2\lambda u} \quad (1) \quad ; \quad y = -\lambda u^2 \Rightarrow dy = -2\lambda u du \quad (1) \\ &= - \int_0^{-\lambda x^2} e^y dy \quad (1) \\ &= -(e^y \Big|_0^{-\lambda x^2}) \quad (1) \\ &= -(e^{-\lambda x^2} - 1) \\ &= 1 - e^{-\lambda x^2} \quad (1) \end{aligned}$$

Question 2: A transmitter produced 30% of radioactive signals with source noise and 70% of good signals that may affect by external noise. 20% of the source bad signals can be managed using a good receiver. If the proportion of signals that can be managed out of all signals transmitted away from the transmitter is 60%, find the proportion of signals that could be managed using the receiver such that the original signals were good when transmitted from the transmitter. (5 marks)



$$0.6 = P(M) = 0.06 + 0.7x \quad (2)$$

$$\Rightarrow x = P(M | \bar{N}) = \frac{0.54}{0.70} = 0.77. \quad (1)$$

Part B: Multiple choices

(16 marks)

Complete the following table by putting the correct symbol in the correct place.

Question	1	2	3	4	5	6	7	8
Answer	b	e	a	e	a	e	d	b

Q 1: Let A and B be two events such that $P(A) = 4/7$, $P(A \cap B) = 3/7$ and $P(A|B) = 2/3$. Then $P(A|\bar{B})$ equals

- a) 0.1 b) 0.4 c) 0.2 d) 0.3 e) 0.5

Q 2: A telecommunication company has noticed that 80% of their signals have no errors when they reach some end point, 15% of their signals have one error, and 5% have two errors. If you buy a new service form this company, the expected number of errors in this service will be

- a) 0.15 b) 0.2 c) 0.8 d) 0.35 e) 0.25

Q 3: Let (X, Y) be a bivariate random variable such that $Var(X) = 9$, $Var(Y) = 4$ and $Cov(X, Y) = 3$, then $Corr(1 - 3X, 3Y - 1) =$

- a) -1/2 b) -1/4 c) 1/2 d) 1/4 e) 1

Q 4: Let X be a discrete random variable with pdf $f(x) = 0.25$, $x = 1, 2, 3, 4$, and zero otherwise, then $E(X) =$

- a) 1 b) 2 c) 1.5 d) 0.25 e) 2.5

For questions 5-6, Consider the following joint p.d.f. of X and Y :

$X \backslash Y$	0	1	2	3
0	1/8	1/8	0	0
1	0	2/8	2/8	0
2	0	0	1/8	1/8

Q 5: The probability that X is greater than Y ($P(X > Y)$) equals

- a) 0 b) 1/8 c) 2/8 d) 3/8 e) 4/8

Q 6: $P(0 \leq X \leq 1 | Y = 1) =$

- a) 0 b) 2/8 c) 4/8 d) 6/8 e) 1

Q 7: In the experiment of tossing a coin two times, the value of the cumulative distribution function (c.d.f.) at the interval $1 \leq x < 2$ equals

- a) 0 b) 1/4 c) 2/4 d) 3/4 e) 1

Q 8: Let A and B be events such that $P(A) = 0.45$, $P(B) = 0.5$ and $P(\bar{A} \cap \bar{B}) = 0.2$. Compute $P(A \cap B)$.

- a) 0.8 b) 0.15 c) 0.95 d) 0.23 e) None

Question 3: Each of two transmitters worked for 100 hours with different wave lengths. Three receivers are supposed to receive their signals based on their wave lengths. The following table shows the connection times (in hours) between the transmitters and the receivers. (6 marks)

xi		Transmitters		Total
		T1	T2	
1	Receiver 1	15	50	65
2	Receiver 2	60	30	90
3	Receiver 3	25	20	45
	Total	100	100	200

- a) Find the proportion of times for each receiver such that the signals were produced from the transmitters T1 and T2.
 b) Find the conditional p.d.f.'s $f_X(x|T1)$ and $f_X(x|T2)$.
 c) Find the marginal p.d.f. $f_X(x)$.

$$a) P(X=1|T_1) = 0.15$$

$$P(X=1|T_2) = 0.5$$

$$P(X=2|T_1) = 0.6$$

$$P(X=2|T_2) = 0.3$$

$$P(X=3|T_1) = 0.25$$

$$P(X=3|T_2) = 0.2$$

$$b) f_X(x|T_1) = 0.15 \delta(x-1) + 0.6 \delta(x-2) + 0.25 \delta(x-3)$$

$$f_X(x|T_2) = 0.5 \delta(x-1) + 0.3 \delta(x-2) + 0.2 \delta(x-3)$$

$$c) P(X=1) = P(X=1|T_1)P(T_1) + P(X=1|T_2)P(T_2)$$

$$= (0.15)(0.5) + (0.5)(0.5) = 0.325$$

$$P(X=2) = (0.6)(0.5) + (0.3)(0.5) = 0.45$$

$$P(X=3) = (0.25)(0.5) + (0.2)(0.5) = 0.225$$

$$f_X(x) = 0.325 \delta(x-1) + 0.45 \delta(x-2) + 0.225 \delta(x-3)$$

$$\text{or } P(X=1) = \frac{65}{200}$$

$$P(X=2) = \frac{90}{200}$$

$$P(X=3) = \frac{45}{200}$$

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Question 4: Let the r.v. X follows the exponential distribution with p.d.f. $f(x) = \begin{cases} \lambda e^{-\lambda x}, & 0 < x < \infty \\ 0, & \text{o.w.} \end{cases}$ (5 marks)

a) Find λ . b) Find $E(X)$, $Var(X)$.

a) $\lambda \int_0^{\infty} e^{-\lambda x} dx = 1$ ---- λ is arbitrary (1)

b) $E(X) = \lambda \int_0^{\infty} x e^{-\lambda x} dx$
 $= \lambda \left(-\frac{x}{\lambda} e^{-\lambda x} - \frac{1}{\lambda^2} e^{-\lambda x} \right) \Big|_0^{\infty} = \frac{1}{\lambda}$ (2)

$E(X^2) = \lambda \int_0^{\infty} x^2 e^{-\lambda x} dx$
 $= \lambda \left(-\frac{x^2}{\lambda} e^{-\lambda x} - \frac{2}{\lambda} x e^{-\lambda x} - \frac{2}{\lambda^2} e^{-\lambda x} \right) \Big|_0^{\infty}$
 $= \lambda \left(\frac{2}{\lambda^2} \right) = \frac{2}{\lambda^2}$ (1)

$Var(X) = E(X^2) - (E(X))^2 = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{2\lambda - 1}{\lambda^2}$ (1)

Question 5: Consider the joint p.d.f. $f(x, y) = \begin{cases} \frac{5}{16} x^2 y, & 0 < y < x < 2 \\ 0, & \text{o.w.} \end{cases}$ (5 marks)

Find the marginal p.d.f.'s for X and Y and then discuss the independence of the random variables.

$f_X(x) = \int_0^2 \frac{5}{16} x^2 y dy = \frac{5}{16} x^2 \left[\frac{y^2}{2} \right]_0^2$
 $= \dots = \frac{(4)(5)}{(16)(2)} x^2$ (1.5)

$f_Y(y) = \int_0^2 \frac{5}{16} x^2 y dx = \frac{5}{16} y \left[\frac{x^3}{3} \right]_0^2$
 $= \dots = \frac{(5)(8)}{(16)(3)} y$ (1.5)

$f(x, y) \neq f_X(x) f_Y(y)$ since

$\frac{5}{16} x^2 y \neq \frac{(4)(5)}{(16)(2)} x^2 \frac{(5)(8)}{(16)(3)} y = \frac{5}{3} \frac{5}{16} x^2 y$

So, not indep. (1)

	1	2	3	4	5	6	7	8
Q1: Ld A,...	b	e	a	e	a	e	d	b
For questions	a	e	d	b	b	e	a	e

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Count -- ,

